CSE 150A-250A AI: Probabilistic Methods

Hw2
Project released.

Lecture 5

Fall 2026

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof Berg-Kirkpatrick)

Agenda

Review

Inference

Exact Inference: Variable Elimination

Review

d-separation and conditional independence

- Theorem

d-separation and conditional independence

- **Theorem**

  $P(X, Y|E) = P(X|E)\,P(Y|E)$ if and only if every *path* from a node in X to a node in Y is *blocked* by E.

d-separation and conditional independence

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d-separation and conditional independence

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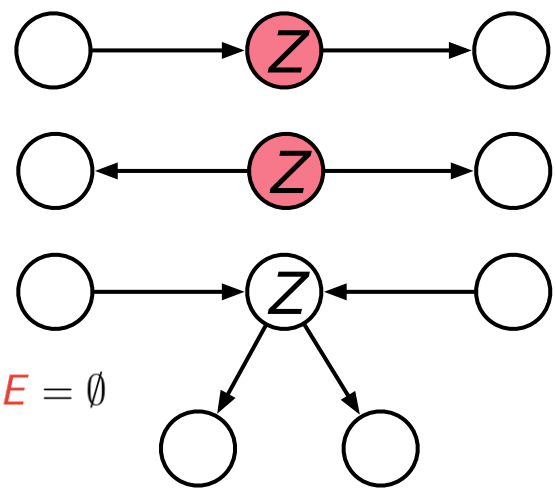
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(1) $Z \in E$ — **causal chain**

(2) $Z \in E$ — **common cause**

(3) $Z \notin E$ — **unobserved common effect**

$\text{descendants}(Z) \cap E = \emptyset$



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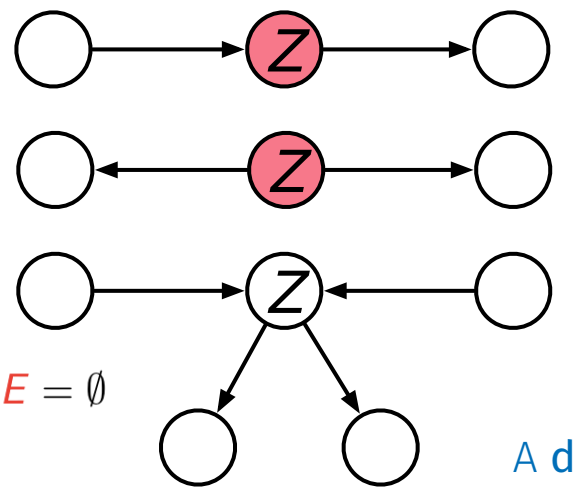
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$\text{descendants}(Z) \cap E = \emptyset$

A **descendant** of Z is *any* node (e.g., child, grandchild) that lies on a directed path from Z.

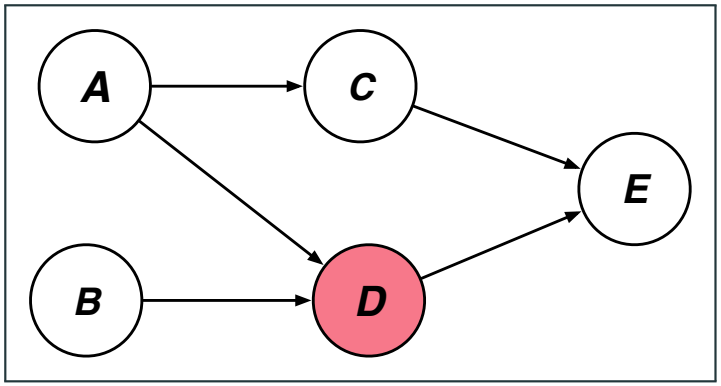
D-Separation Algorithm

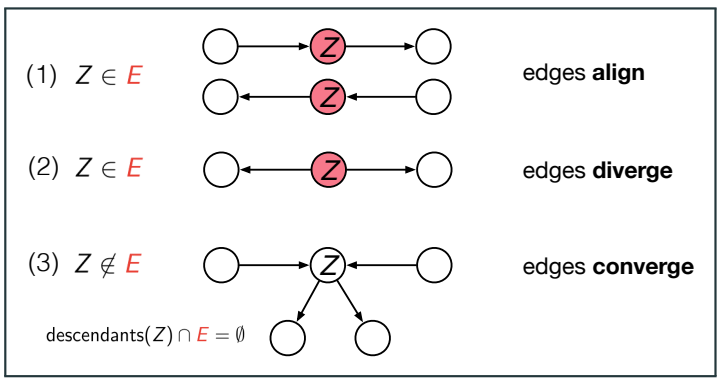
1. Shade all observed nodes $\{Z_1, \ldots, Z_k\}$ in the graph.
2. Enumerate all undirected paths from X to Y.
3. For each path:
   - 3.1 Decompose the path into triples (segments of 3 nodes).
   - 3.2 If none of the d-separation blocking conditions apply to any of the triples on the path, then the path is **active** and **d-connects** X and Y. Return $X \not\perp\!\!\!\perp Y \mid \{Z_1, \ldots, Z_k\}$
4. If all paths are blocked , then

$$X \perp\!\!\!\perp Y \mid \{Z_1, \ldots, Z_k\}.$$

Loopy example

A. TRUE or B. FALSE?

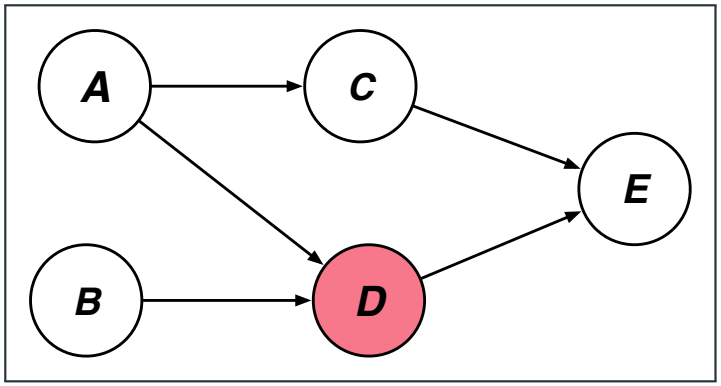


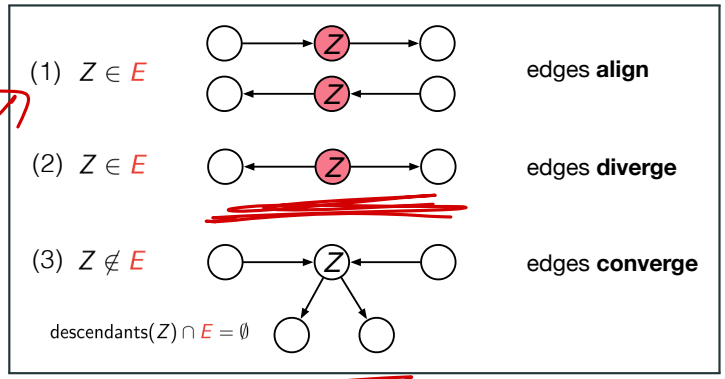


Loopy example

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5. $P(B|D,E) \stackrel{?}{=} P(B|D)$



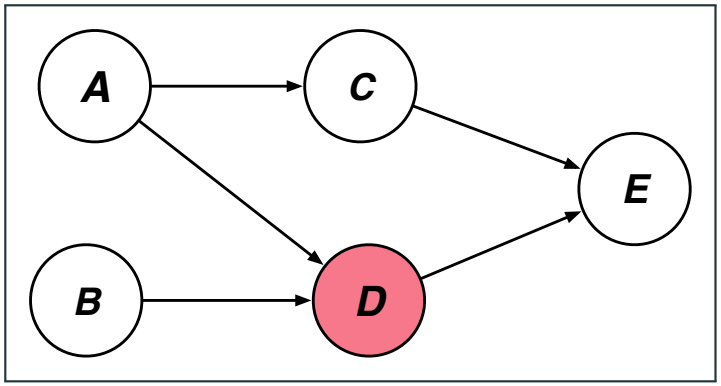


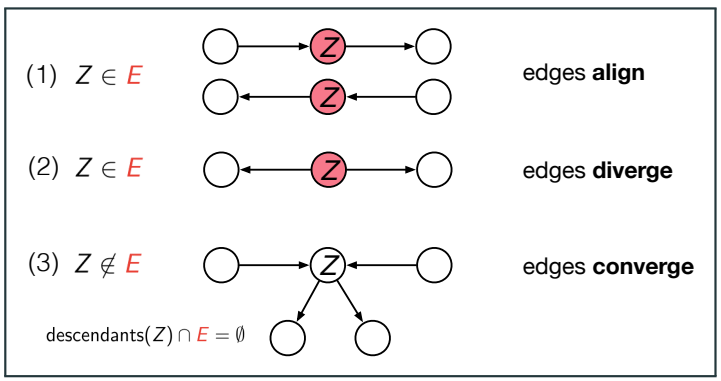
Loopy example

A. TRUE or B. FALSE?

5. $P(B|D,E) \stackrel{?}{=} P(B|D)$

   The evidence is $\{D\}$.





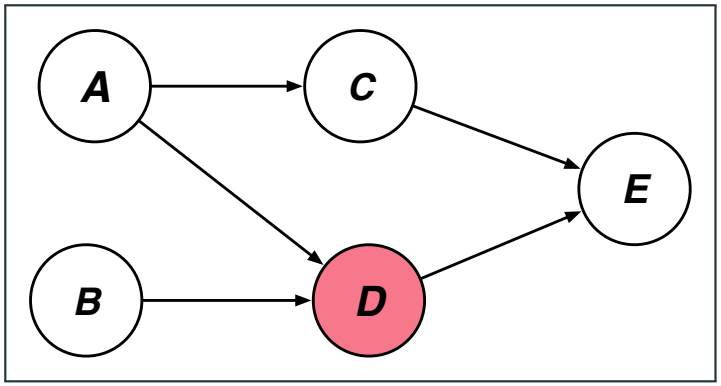
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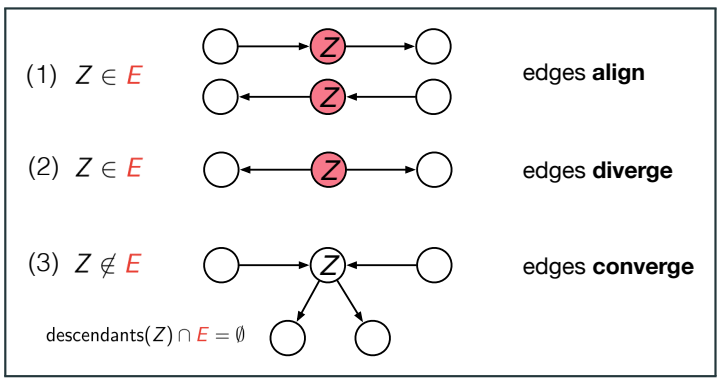
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5. $P(B|D,E) \stackrel{?}{=} P(B|D)$

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   There are two paths from B to E.





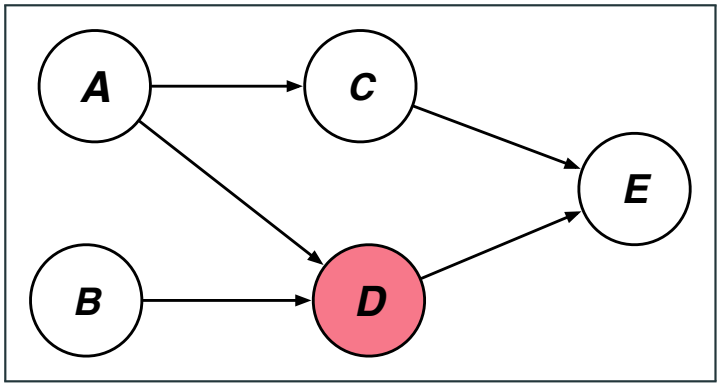
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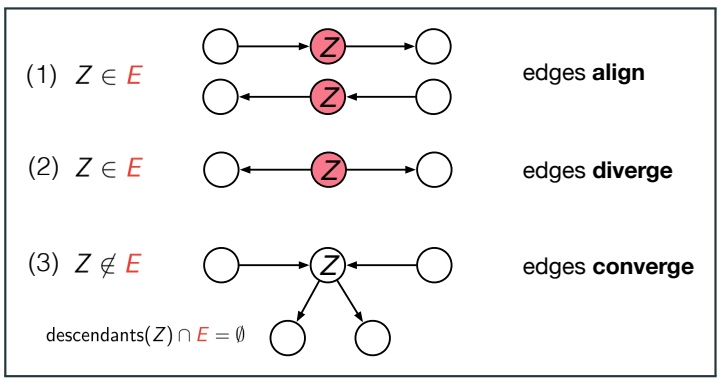
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5. $P(B|D,E) \stackrel{?}{=} P(B|D)$

   The evidence is $\{D\}$.
   There are two paths from B to E.

   Path $B \to D \to E$
   is blocked by node D,
   satisfying condition (1).





Loopy example

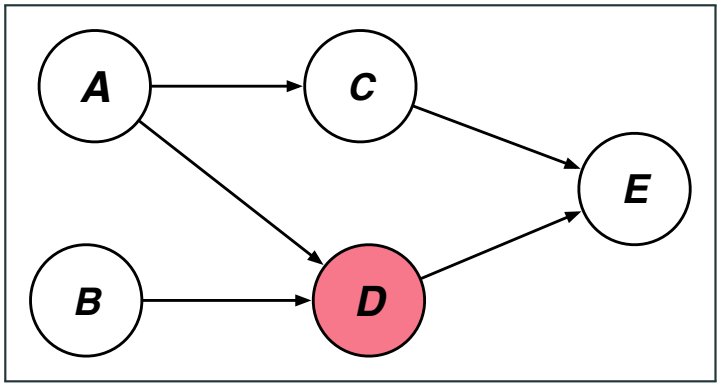
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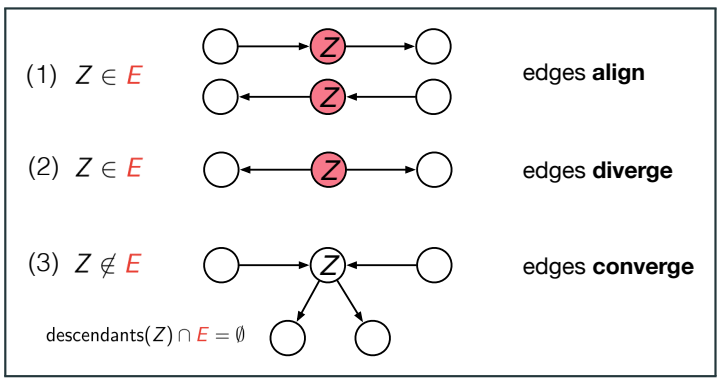
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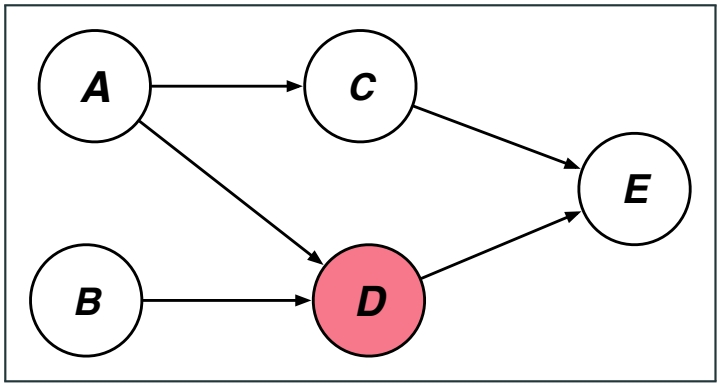
   Path $B \to D \leftarrow A \to C \to E$
   is not blocked by any node.





Loopy example

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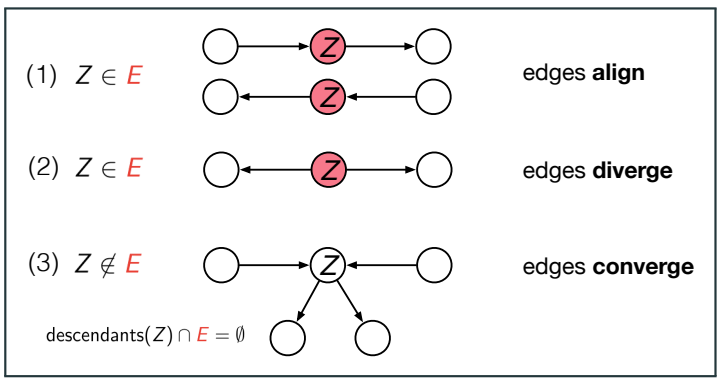
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   The statement is **false**.

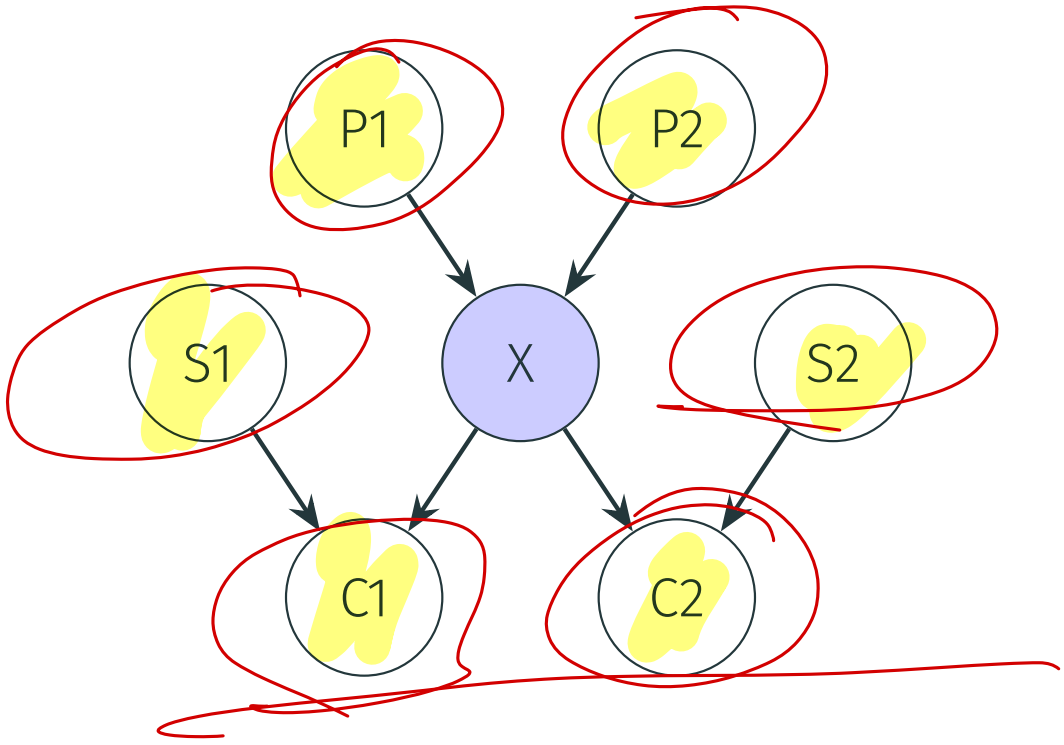


Markov Blanket

A Markov Blanket B_X of node X consists of **parents** of X, **children** of X and **"spouses"** (other parents of children of X, but not X) of X.

Markov Blanket

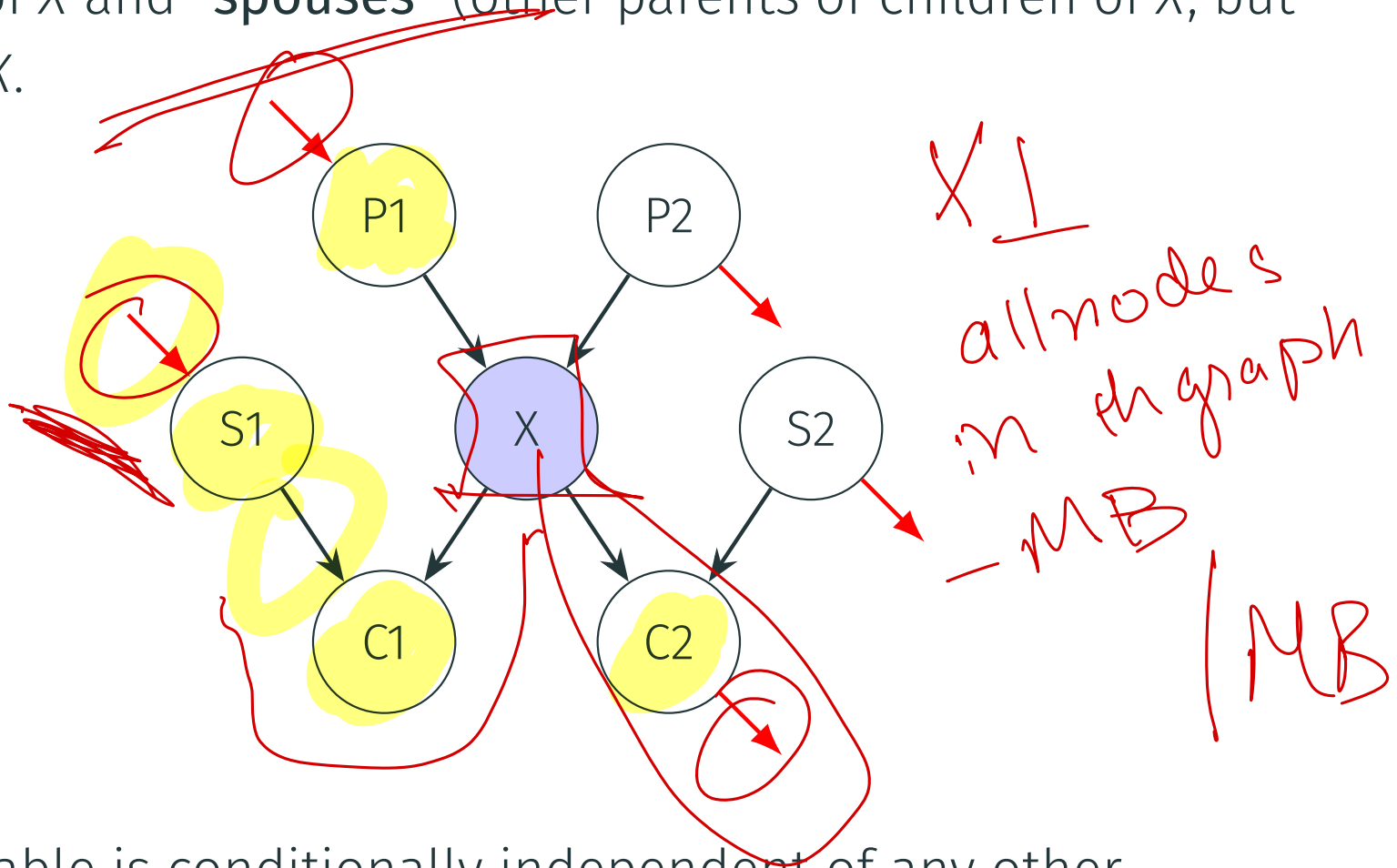
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  How to express $P(Q|E)$ in terms of the CPTs $P(X_i|pa(X_i))$ of the BN, which are assumed to be given?

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- **Tools at our disposal**

  Bayes rule
  marginalization
  product rule
  marginal independence
  conditional independence

Strategy to compute $P(Q|E)$

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Bayes rule

Use to express $P(Q|E)$ in terms of conditional probabilities that respect the order of the DAG.

Strategy to compute $P(Q|E)$

Bayes rule	Use to express $P(Q\|E)$ in terms of conditional probabilities that respect the order of the DAG.
marginalization	Use to introduce nodes on the left side of the conditioning bar when they need to appear as parents.

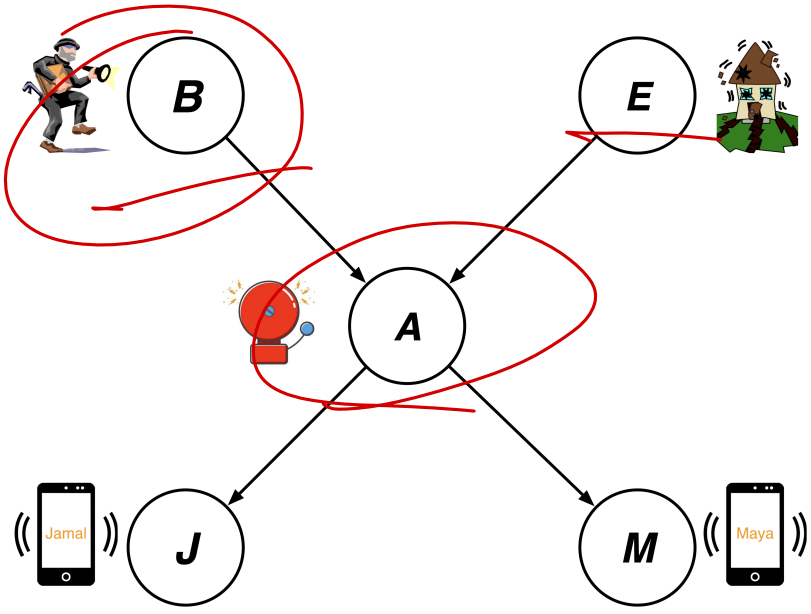
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marginalization	Use to introduce nodes on the left side of the conditioning bar when they need to appear as parents.
product rule	Use to express joint predictions (over multiple variables) in terms of simpler individual predictions.
marginal and conditional independence	Use to remove non-informative variables from the right side of the conditioning bar.

Inference Example



CPT ?

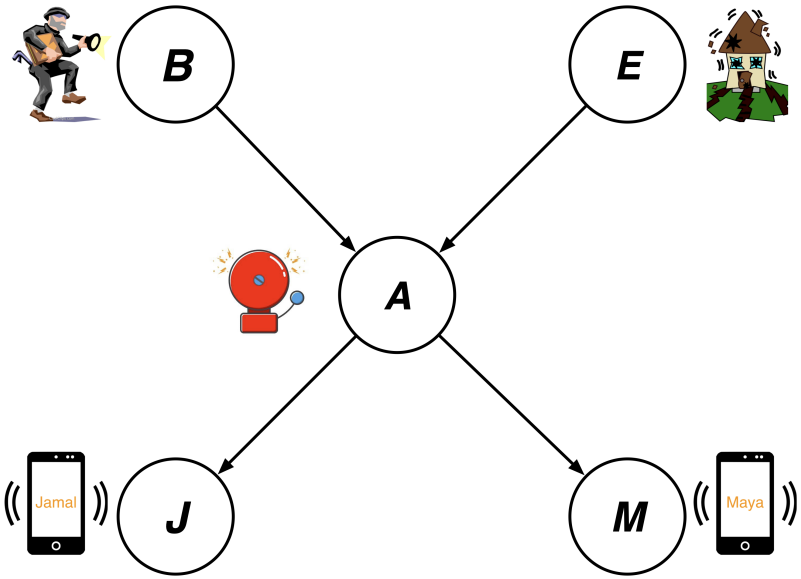
$\rightarrow P(X \mid Pa(X))$

Q. What are the CPTs associated with the DAG shown above?

A. P(B|J,M) **B.** P(B) **C.** P(A|B,E)

D. A, B and C **E.** B and C

Inference Example

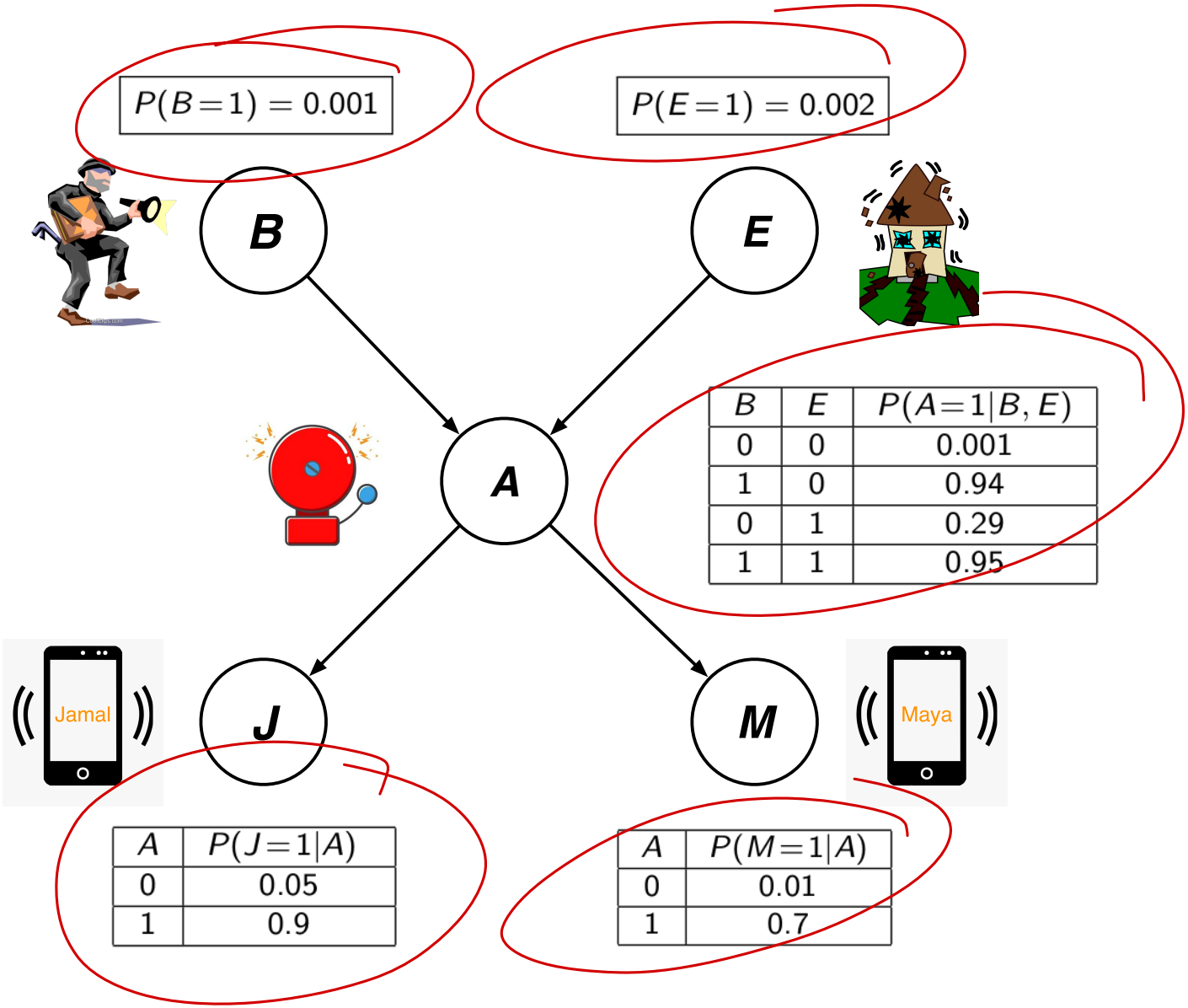


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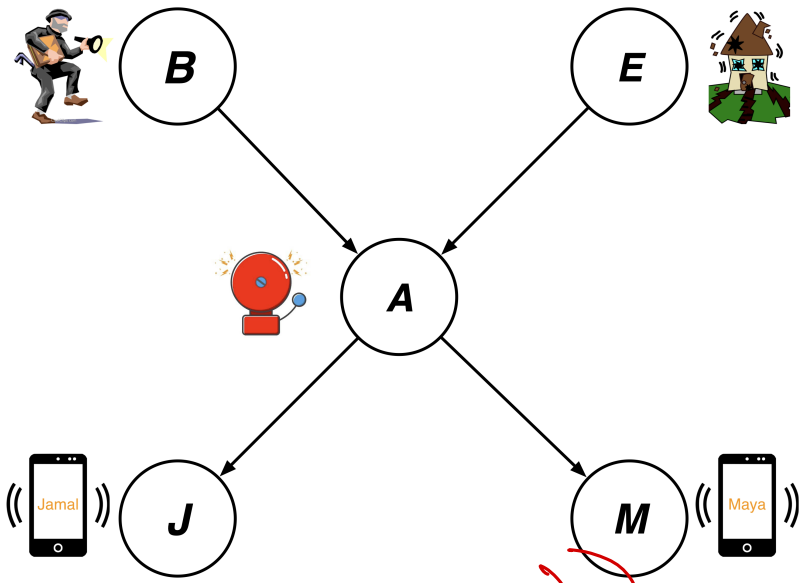


B	E	$P(A=1\|B,E)$
0	0	0.001
1	0	0.94
0	1	0.29
1	1	0.95

A	$P(J=1\|A)$
0	0.05
1	0.9

A	$P(M=1\|A)$
0	0.01
1	0.7

Inference Example



$$P(B = 1 | J = 1, M = 1) = ??$$

$$P(B=1 \mid J=1, M=1) = \frac{P(J=1, M=1 \mid B=1)\, P(B=1)}{P(J=1, M=1)}$$

Inference Example: Enumeration

$$P(B = 1 | J = 1, M = 1)$$

$$= \frac{P(B = 1, J = 1, M = 1)}{P(J = 1, M = 1)}$$

Bayes + product.

convert this to the full joint.

$\frac{1}{\alpha}$

A and E

$$\sum_a \sum_e P(B=1, J=1, M=1, A=a, E=e)$$

Inference Example: Enumeration

$$P(B = 1 | J = 1, M = 1)$$

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$$= \alpha \sum_e \sum_a P(B = 1) P(E = e) P(A = a | B = 1, E = e) P(J = 1 | A = a) P(M = 1 | A = a)$$

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$$= \alpha \sum_e \sum_a P(B = 1) P(E = e) P(A = a | B = 1, E = e) P(J = 1 | A = a) P(M = 1 | A = a)$$

Q. How many operations does it take to calculate $P(B = 1, J = 1, M = 1)$?

Inference Example: Enumeration

$$P(B = 1, J = 1, M = 1)$$

$$\begin{aligned}
&= P(B = 1) \times P(E = 0) \times P(A = 0|B = 1, E = 0) \times P(J = 1|A = 0) \times P(M = 1|A = 0) \\
&+P(B = 1) \times P(E = 0) \times P(A = 1|B = 1, E = 0) \times P(J = 1|A = 1) \times P(M = 1|A = 1) \\
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\end{aligned}$$

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\end{aligned}$$

A. 16 multiplications and 3 additions.

Inference Example: Enumeration

$$P(B = 1 \mid J = 1, M = 1) = \frac{P(B = 1, J = 1, M = 1)}{P(B = 0, J = 1, M = 1) + P(B = 1, J = 1, M = 1)}$$

Q. How many multiplications and additions does it take to compute $P(B = 1 \mid J = 1, M = 1)$?

A. 16 multiplications, 3 additions

B. 48 multiplications, 7 additions

C. 32 multiplications, 6 additions

D. 32 multiplications, 7 additions

E. I have no idea

Inference Example: Enumeration

$$P(B = 1 | J = 1, M = 1)$$

$$= \frac{P(B = 1, J = 1, M = 1)}{P(J = 1, M = 1)}$$

$$= \alpha P(B = 1, J = 1, M = 1)$$

$$= \alpha \sum_e \sum_a P(B = 1, J = 1, M = 1, E = e, A = a)$$

$$= \alpha \sum_e \sum_a P(B = 1) P(E = e) P(A = a | B = 1, E = e) P(J = 1 | A = a) P(M = 1 | A = a)$$

$$= \alpha P(B = 1) \sum_e P(E = e) \sum_a P(A = a | B = 1, E = e) P(J = 1 | A = a) P(M = 1 | A = a)$$

Inference Example: Enumeration

$$P(B = 1 | J = 1, M = 1)$$

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$$= \alpha \sum_e \sum_a P(B = 1, J = 1, M = 1, E = e, A = a)$$

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$$= \alpha P(B = 1) \sum_e P(E = e) \sum_a P(A = a | B = 1, E = e) P(J = 1 | A = a) P(M = 1 | A = a)$$

Q. How many operations does it take now to calculate $P(B = 1, J = 1, M = 1)$?

Inference Example: Enumeration

$$P(B = 1, J = 1, M = 1)$$

$$\begin{aligned}
= P(B = 1) \times \Big[& P(E = 0) \times \big(P(A = 0 | B = 1, E = 0) \times P(J = 1 | A = 0) \times P(M = 1 | A = 0) \\
& + P(A = 1 | B = 1, E = 0) \times P(J = 1 | A = 1) \times P(M = 1 | A = 1) \big) \\
+ P(E = 1) \times \big(& P(A = 0 | B = 1, E = 1) \times P(J = 1 | A = 0) \times P(M = 1 | A = 0) \\
& + P(A = 1 | B = 1, E = 1) \times P(J = 1 | A = 1) \times P(M = 1 | A = 1) \big) \Big]
\end{aligned}$$

Inference Example: Enumeration

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& + P(A = 1|B = 1, E = 0) \times P(J = 1|A = 1) \times P(M = 1|A = 1) \big) \\
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& + P(A = 1|B = 1, E = 1) \times P(J = 1|A = 1) \times P(M = 1|A = 1) \big) \Big]
\end{aligned}$$

A. 11 multiplications and 3 additions.

Inference Example: Enumeration

$$P(b|j,m) = \alpha P(b) \sum_E P(E) \sum_A P(A|b,E) P(j|A) P(m|A)$$

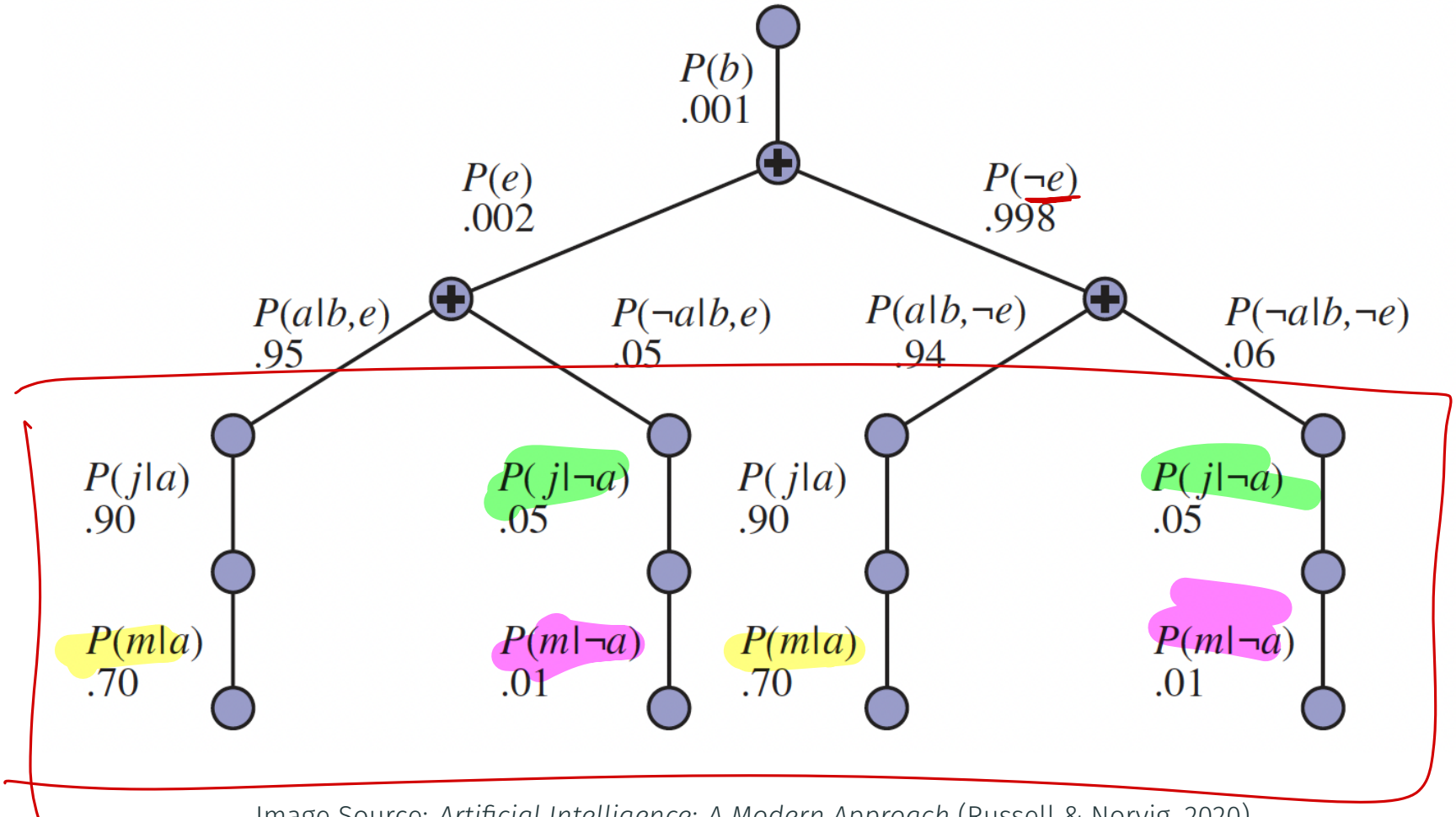


Image Source: *Artificial Intelligence: A Modern Approach* (Russell & Norvig, 2020)

Inference Example: Enumeration

$$P(b|j,m) = \alpha P(b) \sum_E P(E) \sum_A P(A|b,E) P(j|A) P(m|A)$$

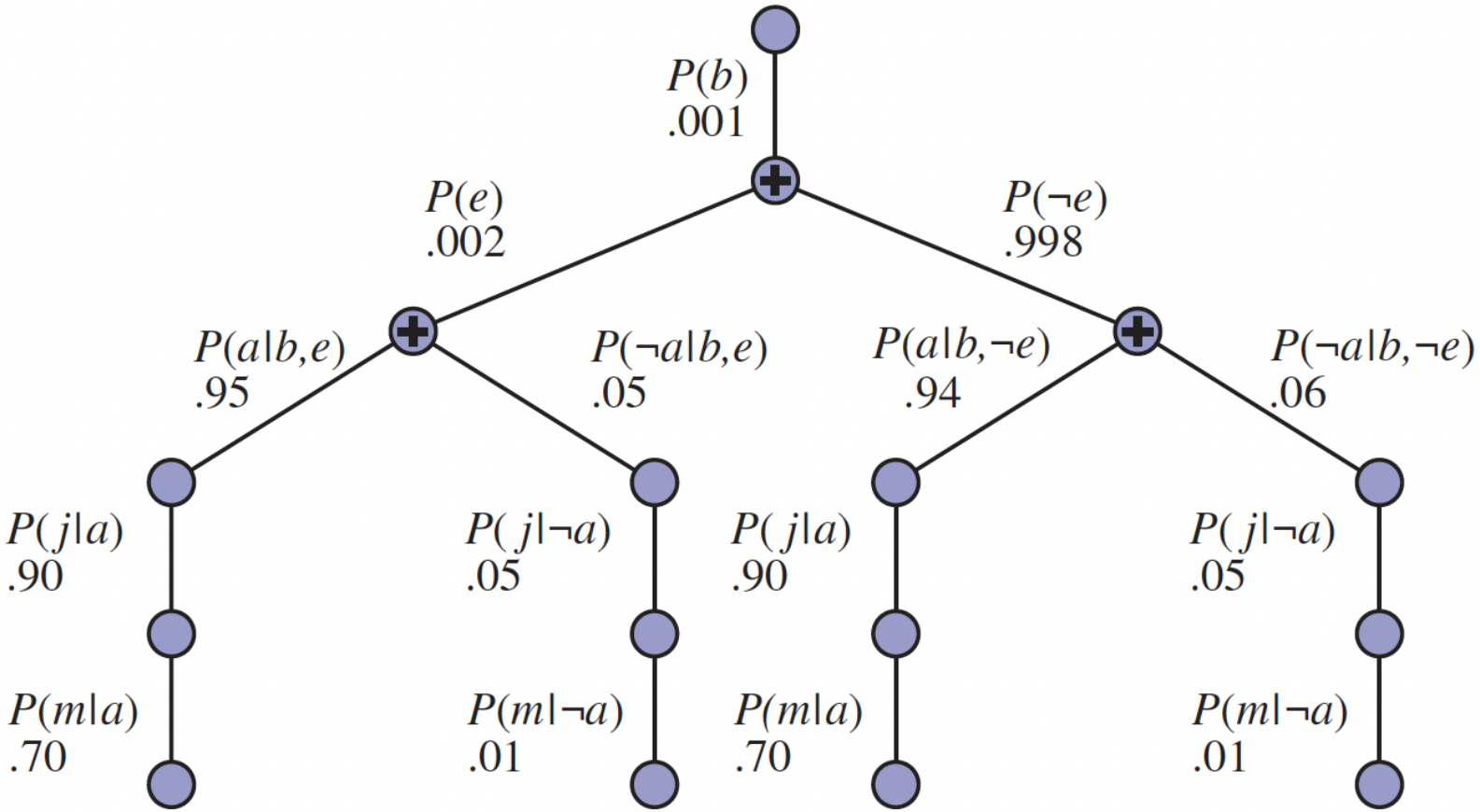


Image Source: *Artificial Intelligence: A Modern Approach* (Russell & Norvig, 2020)

Repeated Computations $\longrightarrow$ Dynamic Programming

Inference Example: Enumeration

$$P(B = 1 \mid J = 1, M = 1) = \frac{P(B = 1, J = 1, M = 1)}{P(B = 0, J = 1, M = 1) + P(B = 1, J = 1, M = 1)}$$

Q. How many multiplications and additions does it take to compute $P(B = 1 \mid J = 1, M = 1)$ using this rearrangement?

A. 11 multiplications, 3 additions

B. 22 multiplications, 6 additions

C. 22 multiplications, 7 additions

D. 33 multiplications, 7 additions

E. I have no idea

Exact Inference: Variable Elimination

Variable Elimination

- **Idea**: Eliminate repeated calculations by storing intermediate results in “factors”.

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- Variable Elimination (VE) successively eliminates all non-query variables, one at a time, until only factors involving the query variables remain.

Variable Elimination

- **Idea**: Eliminate repeated calculations by storing intermediate results in “factors”.
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- To e

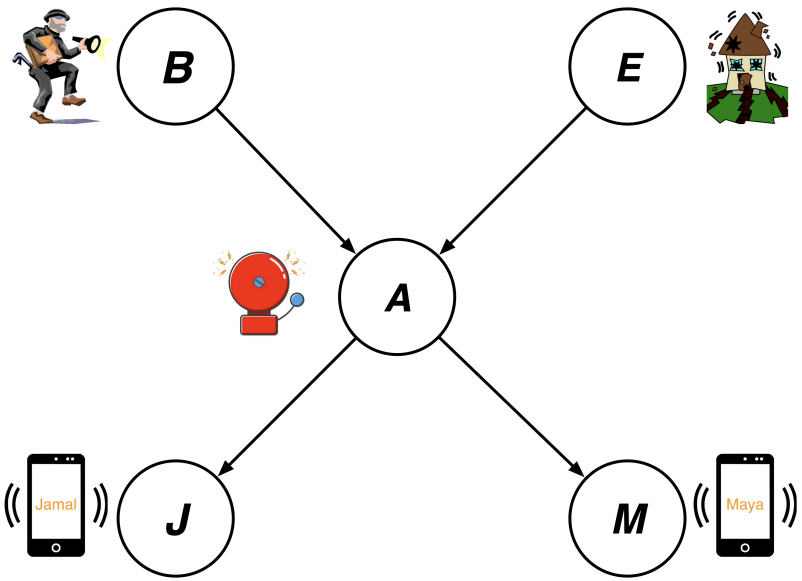
Variable Elimination: Algorithm

To compute the posterior distribution of a query variable given observations:

1. Construct a factor for each conditional probability distribution.
2. Eliminate each non-query variable in some *order*:
   - If the variable is **observed**, set its value to the observed value in every factor containing it.
   - Otherwise, **sum out** the variable:

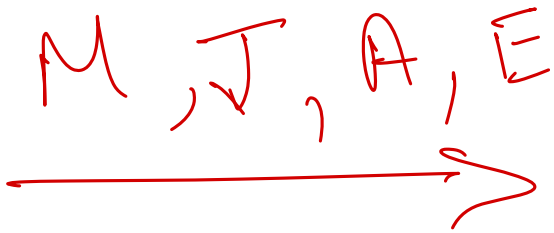
factor.

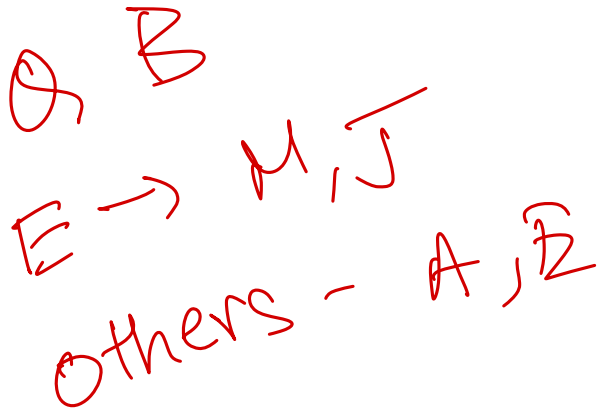
VE Example



$$P(B = 1 | J = 1, M = 1) = ??$$

VE Example

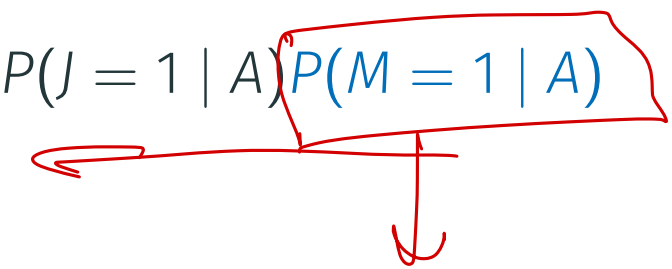


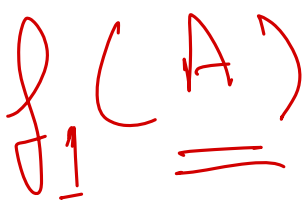


$$P(B \mid J = 1, M = 1)$$

$$= \alpha P(B, J = 1, M = 1)$$

$$= \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) P(J = 1 \mid A) P(M = 1 \mid A)$$





VE Example

$$
\begin{aligned}
&P(B \mid J=1, M=1) \\
&\quad = \alpha P(B, J=1, M=1) \\
&\quad = \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) P(J=1 \mid A) P(M=1 \mid A) \\
&\quad = \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) P(J=1 \mid A) f_1(A)
\end{aligned}
$$

VE Example

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$$= \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) P(J=1 \mid A) P(M=1 \mid A)$$

$$= \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) P(J=1 \mid A) f_1(A)$$

$$= \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) f_2(A)$$

A, B, E

A, B A E

B E

$P(A=1 \mid __) f(A=1)$

$+ P(A=1 \mid __) f(A=0)$

VE Example

$$
\begin{aligned}
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&\quad = \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) P(J=1 \mid A) f_1(A) \\
&\quad = \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) f_2(A) \\
&\quad = \alpha P(B) \sum_E P(E) f_3(B, E)
\end{aligned}
$$

VE Example

$$
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&\quad = \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) f_2(A) \\
&\quad = \alpha P(B) \sum_E P(E) f_3(B, E) \\
&\quad = \alpha P(B) f_4(B)
\end{aligned}
$$

VE Example

$$P(B \mid J=1, M=1)$$

$$= \alpha P(B, J=1, M=1)$$

$$= \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) P(J=1 \mid A) P(M=1 \mid A)$$

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$$= \alpha P(B) \sum_E P(E) f_3(B, E)$$

$$= \alpha P(B) f_4(B)$$

$$= \alpha f_5(B)$$

$$P(A=1 \mid B, E) \cdot f_2(A=1) + P(A=0 \mid B, E) \cdot f_2(A=0)$$

VE Example

$$P(B \mid J = 1, M = 1) = \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) P(J = 1 \mid A) P(M = 1 \mid A)$$

A	$f_1(A) = P(M = 1 \mid A)$
0	0.01
1	0.7

VE Example

$P(B \mid J=1, M=1) = \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) P(J=1 \mid A) f_1(A)$

A	$P(J=1 \mid A)$
0	0.05
1	0.9

A	$f_1(A)$
0	0.01
1	0.7

A	$f_2(A) = P(J=1 \mid A) f_1(A)$
0	0.05 x 0.01
1	0.9 x 0.7

VE Example

$P(B \mid J=1, M=1) = \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) P(J=1 \mid A) f_1(A)$

A	$P(J=1 \mid A)$
0	0.05
1	0.9

A	$f_1(A)$
0	0.01
1	0.7

A	$f_2(A) = P(J=1 \mid A) f_1(A)$
0	0.05×0.01
1	0.9×0.7

VE Example

$$P(B \mid J=1, M=1) = \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) f_2(A)$$

A	$f_2(A)$
0	0.0005
1	0.63

B	E	$P(A=1 \mid B, E)$
0	0	0.001
1	0	0.94
0	1	0.29
1	1	0.95

B	E	$f_3(B,E) = \sum_A P(A \mid B, E) f_2(A)$
0	0	.001 × 0.63 + 0.— × 0.000
1	0	
0	1	
1	1	

$P(A=0 \mid B,E=0)$

$f_2(A=0)$

$P(A=1 \mid B=0, E=0)$

$f_2(A=1)$

VE Example

$$P(B \mid J=1, M=1) = \alpha P(B) \sum_E P(E) \sum_A P(A \mid B, E) f_2(A)$$

A	$f_2(A)$
0	0.0005
1	0.63

B	E	$P(A=1 \mid B, E)$
0	0	0.001
1	0	0.94
0	1	0.29
1	1	0.95

B	E	$f_3(B, E) = \sum_A P(A \mid B, E) f_2(A)$
0	0	$0.001 \times 0.63 + 0.999 \times 0.0005$
1	0	$0.94 \times 0.63 + 0.06 \times 0.0005$
0	1	$0.29 \times 0.63 + 0.71 \times 0.0005$
1	1	$0.95 \times 0.63 + 0.05 \times 0.0005$

VE Example

$$P(B \mid J = 1, M = 1) = \alpha P(B) \sum_E P(E) f_3(B, E)$$

E	$P(E)$
0	0.998
1	0.002

B	E	$f_3(B, E)$
0	0	0.0011295
1	0	0.59223
0	1	0.183055
1	1	0.598525

B	$f_4(B) = \sum_E P(E) f_3(B, E)$
0	$0.998 \times 0.0011295 + 0.002 \times 0.183055$
1	$0.998 \times 0.59223 + 0.002 \times 0.598525$

VE Example

$$P(B \mid J = 1, M = 1) = \alpha P(B) f_4(B)$$

B	$P(B)$
0	0.999
1	0.001

B	$f_4(B)$
0	0.001493351
1	0.59224259

B	$f_5(B) = P(B)f_4(B)$
0	0.999×0.001493351
1	0.001×0.59224259

VE Example

$$P(B \mid J = 1, M = 1) = \alpha f_5(B)$$

B	$f_5(B)$
0	0.001491858
1	0.000592243

divide that by their sum

Normalization

VE Example

$$P(B \mid J = 1, M = 1) = \alpha f_5(B)$$

B	$f_5(B)$
0	0.001491858
1	0.000592243

$$N = 0.001491858 + 0.000592243, \qquad \alpha = \frac{1}{N}$$

B	$P(B \mid J = 1, M = 1)$
0	0.716
1	0.284

VE Example

$$P(B \mid J = 1, M = 1) = \alpha f_5(B)$$

B	$f_5(B)$
0	0.001491858
1	0.000592243

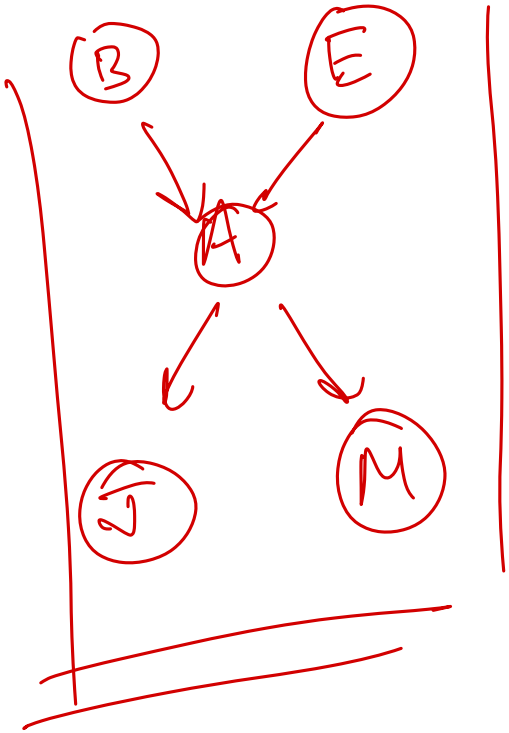
$$N = 0.001491858 + 0.000592243, \qquad \alpha = \frac{1}{N}$$

B	$P(B \mid J = 1, M = 1)$
0	0.716
1	0.284

ignore divisions

Q. How many multiplications and additions does it take to compute $P(B = 1 \mid J = 1, M = 1)$ using VE?

VE Example



$$f_1(A) = P(M = 1 \mid A)$$

$$f_2(A) = P(J = 1 \mid A) f_1(A)$$

$$f_3(B, E) = \sum_A P(A \mid B, E) f_2(A)$$

$$f_4(B) = \sum_E P(E) f_3(B, E)$$

$$f_5(B) = P(B) f_4(B)$$

$$P(B \mid J = 1, M = 1) = \alpha f_5(B)$$

VE Example

Factor	Multiplications	Additions
$f_1(A)$	0	0
$f_2(A)$	2	0
$f_3(B, E)$	8	4
$f_4(B)$	4	2
$f_5(B)$	2	0
Total	**16**	**6**

Normalization: $+1$ addition

16 multiplications, 7 additions

VE: Factors

- Factors are usually represented as a table (therefore an arbitrary function)
- Caution: Factors can look like CPTs, and CPTs can be represented as factors, but factors are **not** necessarily probabilities!
- The values in factors only represent intermediate values in the calculations of some probability - with no real meaning in themselves.

VE: Complexity

Does order of elimination matter?

- In general, yes (but not in the trivial graphs we've been considering)
- Time and space of VE is dominated by the **largest** factor created
- **Heuristic:** Eliminate the variable that will lead to the smallest next factor being created
    - In a **polytree** this leads to **linear** time inference (in size of largest CPT).

Polytree: A belief network with at most one undirected path between any two nodes.

That’s All Folks!